

Week 7

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CSE B

Roll- 64

Questions:

Q1. For given subset of grammar 7.1, design RD parser with appropriate error messages with expected character and row and column number.

Program main() { declarations assign_stat}

declarations → data-type identifier-list; declarations | epsilon

data-type → int | char

identifier-list → id | id, identifier-list

assign_stat → id=id; | id = num;

CODE:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
#include "la.h"
```

```
struct token lookahead;
```

```
FILE *fp;
```

```
void error(char *expected)
```

```
{
```

```
    printf("\nERROR at row %d col %d\n", lookahead.row, lookahead.col);
```

```
    printf("Expected: %s\n", expected);  
    printf("Found : %s (%s)\n", lookahead.lexeme, lookahead.type);  
    exit(0);  
}
```

```
void match(char *lex)  
{  
    if (strcmp(lookahead.lexeme, lex) == 0)  
        lookahead = getNextToken(fp);  
    else  
        error(lex);  
}
```

```
void matchType(char *type)  
{  
    if (strcmp(lookahead.type, type) == 0)  
        lookahead = getNextToken(fp);  
    else  
        error(type);  
}
```

```
void Program();  
void declarations();  
void datatype();  
void identifier_list();
```

```
void assign_stat();
```

```
void Program()
```

```
{  
    if (strcmp(lookahead.lexeme, "main") == 0)  
    {  
        match("main");  
        match("(");  
        match(")");  
        match("{");  
        declarations();  
        assign_stat();  
        match("}");  
        printf("\nSUCCESS: Valid Program\n");  
    }  
    else  
        error("main");  
}
```

```
void declarations()
```

```
{  
    if (strcmp(lookahead.lexeme, "int") == 0 ||  
        strcmp(lookahead.lexeme, "char") == 0)  
    {  
        datatype();  
        identifier_list();  
    }
```

```

    match(";");
    declarations(); // recursion
}
// else epsilon
}

void datatype()
{
    if (strcmp(lookahead.lexeme, "int") == 0)
        match("int");
    else if (strcmp(lookahead.lexeme, "char") == 0)
        match("char");
    else
        error("int or char");
}

void identifier_list()
{
    matchType("Identifier");

    if (strcmp(lookahead.lexeme, ",") == 0)
    {
        match(",");
        identifier_list();
    }
}

```

```
void assign_stat()
{
    matchType("Identifier");
    match("=");

    if (strcmp(lookahead.type, "Identifier") == 0)
        matchType("Identifier");
    else if (strcmp(lookahead.type, "Number") == 0)
        matchType("Number");
    else
        error("Identifier or Number");

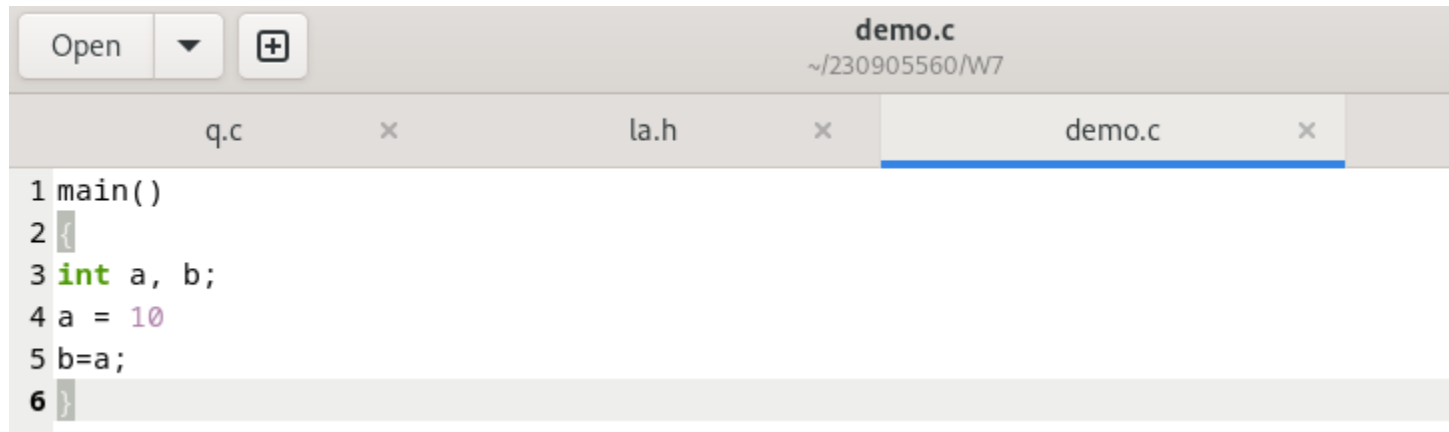
    match(";");
}
```

```
int main()
{
    fp = fopen("demo.c", "r");

    if (fp == NULL)
    {
        printf("Cannot open file\n");
        return 0;
    }
}
```

```
    lookahead = getNextToken(fp);  
    Program();  
  
    fclose(fp);  
    return 0;  
}
```

Sample input:



The screenshot shows a code editor window titled "demo.c" with the path "~/230905560/W7". The editor has three tabs: "q.c", "la.h", and "demo.c". The "demo.c" tab is active and shows the following code:

```
1 main()  
2 {  
3 int a, b;  
4 a = 10  
5 b=a;  
6 }
```

Output:



The screenshot shows a terminal window with the following output:

```
cd1-6cse-b2@sce-cl11-30:~/230905560/W7$ ./q  
  
ERROR at row 5 col 1  
Expected: ;  
Found   : b (Identifier)  
cd1-6cse-b2@sce-cl11-30:~/230905560/W7$
```